Write a Program in C to implement quick Sort.

①

```
# include <stdio.h>
void quicksort (int number[25], int first, int last)
{
int i, j, pivot, temp;
if (first < last)
{
pivot = first;
i = first;
j = last;
while (i < j)
{
while (number[i] <= number [pivot] && i < last)
i++;
while (number[j] > number [pivot])
j--;
if (i<j)
{
temp = number [i];
```

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```
number[i] = number[j];
number[j] = temp;

}
}

temp = number[pivot];
number[pivot] = number[j];
number[j] = temp;
quicksort (number, first, j-1);
quicksort (number, j+1, last);
}
}

int main()
{
int i, count, number[25];
Printf ("How many elements are u going to
        enter? : ");
scanf ("%d", &count);
Printf ("Enter % elements:", count);
for (i=0; i<count; i++)
scanf ("%d", &number[i]);
quicksort (number, 0, count-1);
```

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```
printf(" order of sorted elements:");
for (i=0; i<count; i++)
printf(" %d", number[i]);
return 0;
}
```

Output:→

How many elements are u going to enter? : 10

Enter 10 elements: 2 3 5 7 1 9 3 8 0 4

Order of sorted elements: 0 1 2 3 4 5 7 8 9

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Write a program to implement Quick Sort in C language

```
#include <stdio.h>
/* Function that consider last element as pivot,
Place the pivot at its exact position, and place
Smaller elements to left of pivot and greater
elements to right of pivot */
int Partition (int a[], int start, int end)
{
int pivot = a[end]; // pivot element
int i = (start -1);
For (int j = start; j<=end-1; j++)
{
// if current element is smaller than the
pivot
if (a[j] < pivot)
{
i++; // increment index of smaller element
int t = a[i];
```

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```
a[i] = a[j];
a[j] = t;
}
}
int t = a[i+1];
a[i+1] = a[end];
a[end] = t;
return (i+1);
}
```

//* function to implement quick sort *//

```
void quick(int a[], int start, int end)//* a[]

{
 if (start < end)
 {
 int p = partition (a, start, end); // p is the partitioning index
 quick (a, start, p-1);
 quick (a, p+1, end);
}}
```

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//* function to print an array *//

```
void PrintArr (int a[], int n)
{
 int i;
for (i=0; i<n; i++)
 Printf ("%d", a[i]);
}

int main()
{
 int a[] = {24, 9, 29, 14, 19, 27};
 int n = sizeof (a) / sizeof (a[0]);
Printf (" Before sorting array elements
          are -\n");
PrintArr (a, n);
quick (a, 0, n-1);
Printf ("\n After sorting array elements
          are -\n");
Print Arr (a, n);
 return 0;
}
```

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